9 **(a)** The state of a sub-atomic particle is represented by a wavefunction $\psi(x)$.

Explain

(i) the physical meaning of the square of the wavefunction amplitude $|\psi(x)|^2$;

.....

..... [1]

(ii) how the probability of the particle residing in the region between two positions, x_1 and x_2, can be determined.

.....

..... [1]

(b) An electron is confined in a one-dimensional infinite potential well of width L. The wavefunction of the electron is

$$\psi(x) = \begin{cases} A\sin\dfrac{n\pi x}{L}, & 0 < x < L \\ 0, & x \le 0 \text{ or } x \ge L \end{cases}$$

where n is a positive integer and A the normalization factor.

For a particular value of n, the wavefunction is shown in Fig. 9.1.

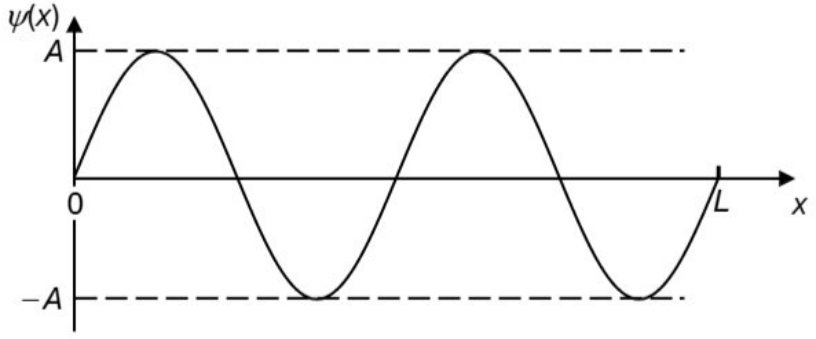


Fig. 9.1

(i) State the value of n for the wavefunction plotted in Fig. 9.1.

n = ... [1]

(ii) Determine an expression for A in terms of L.

A = ... [3]

(iii) Determine the probability of finding the particle between $x = \frac{L}{8}$ and $x = \frac{L}{2}$.

probability = ... [2]

(iv) The one-dimensional well is of length $L = 3.5 \times 10^{-10}$ m.

Determine the energy, in eV, of the quantum state shown in Fig. 9.1.

energy = ... eV [2]

(c) **(i)** A beam of white light passes through a cool, low-pressure gas before it is diffracted by a diffraction grating that is placed perpendicular to the direction of the beam. The resulting beam of light is observed on a screen.

1. State the name of the spectrum observed on the screen.

Describe its appearance.

name: .....

appearance: .....

..... [2]

2. Explain the formation of the spectrum.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(ii) The white light source is removed, and the same gas is heated until it glows.

State the name of the spectrum observed on the screen.

Describe its appearance.

name: .....

appearance: .....

..... [1]

(iii) The ground state energy of the atoms of the cold gas is −13.6 eV, and the first excited state energy is −3.4 eV.

Determine the wavelength of the photon corresponding to the transition between these two energy levels.

State which region of the electromagnetic spectrum this photon belongs to.

wavelength = ... nm

region of EM spectrum = ... [3]

[Total: 20]